

Class 10th Science (Physics, Chemistry, Biology) Syllabus

Theory Marks: 80 | Practical Marks: 20 | Total Marks: 100 | Duration: 3 Hours

Section I: Physics (26 Marks)

Unit 1: Light - Reflection and Refraction (8 Marks)

- Reflection of light by spherical mirrors; image formation; mirror formula and magnification with numericals.
- Refraction through a glass slab, refractive index, conditions for no refraction.
- Spherical lenses: image formation, lens formula, magnification, and power of a lens.

Unit 2: The Human Eye and the Colorful World (5 Marks)

- Structure and accommodation of the human eye; defects of vision (myopia, hypermetropia, presbyopia) and corrections.
- Refraction and dispersion of light through a glass prism.
- Atmospheric refraction: twinkling of stars, advanced sunrise and delayed sunset.

Unit 3: Electricity (6 Marks)

- Electric current, potential difference, Ohm's law and experimental verification.
- Resistance and its dependence on factors; combination of resistors in series and parallel.
- Joule's heating effect of current and applications; electric power and energy calculations.

Unit 4: Magnetic Effects of Current (7 Marks)

- Oersted's experiment; magnetic field and field lines around straight conductor, circular loop, and solenoid.
- Force on a current-carrying conductor in a magnetic field; Fleming's left-hand rule.
- Domestic electric circuits, earthing, short circuit, and safety fuses.

Section II: Chemistry (26 Marks)

Unit 1: Chemical Reactions and Equation (6 Marks)

- Chemical equation, writing of chemical equation; Balancing chemical equations.
- Types of chemical reactions, viz. Combination reactions; Decomposition reactions; Displacement reactions; Double displacement reactions; Oxidation and reduction.
- Effects of oxidation and reduction reactions in everyday life, viz, corrosion and rancidity.

Unit 2: Carbon and its Compounds (8 Marks)

- Bonding in Carbon, Covalent bond, Allotropes of carbon;

- Versatile nature of carbon; Saturated and unsaturated hydrocarbons; chains; branches and rings, homologous series and its characteristics; nomenclature of Carbon compounds.
- Chemical properties of carbon compounds viz. combustion; oxidation; addition and substitution reactions.
- Important Carbon compounds like Ethanol and Ethanoic acid. Properties of Ethanol and Ethanoic acid.
- Soaps and Detergents.

Unit 3: Metals and Non-metals (7 Marks)

- Physical properties of metals and non-metals.
- Chemical properties of metals like action of water, air, acids, salts; Reactivity series of metals.
- Cause of reactivity of metals and non-metals. Properties of Ionic compounds.
- Occurrence of metals; their extraction, enrichment of ores. Extraction of metals in accordance with activity series; refining of metals.
- Corrosion of metals and its prevention.

Unit 4: Acids, Bases and Salts (5 Marks)

- Idea about acids and bases; chemical properties of acids and bases viz. Action of metals, metal carbonates, metal hydrogen carbonates (only in case of acids), metallic acids; non-metallic acids and bases.
- Similarities in acids and bases; reaction of acids and bases with water.
- Strength of acid and base solutions; pH, its importance.
- Idea of salts, their family and pH. Chemicals from common salts like Sodium hydroxide, Baking soda and Washing soda; Hydrated salts Plaster of Paris.

Section III: Biology (28 Marks)

Unit 1: Life Processes (8 Marks)

- Autotrophic and heterotrophic nutrition; human digestive system and nutrition.
- Respiration in plants and animals; aerobic and anaerobic breakdown of glucose.
- Transportation in human beings (heart structure, blood circulation) and plants (xylem & phloem).
- Excretion in human beings (nephron structure and urine formation) and plants.

Unit 2: Control and Co-ordination (6 Marks)

- Nervous system in animals: neuron structure, reflex arc, human brain structure and functions.
- Co-ordination in plants: tropic movements (phototropism, geotropism), plant growth hormones.
- Endocrine glands and hormones in animals (thyroid, pituitary, adrenal, pancreas).

Unit 3: How do Organisms Reproduce (6 Marks)

- Asexual reproduction: fission, fragmentation, regeneration, budding, vegetative propagation, spore formation.

- Sexual reproduction in flowering plants: flower parts, pollination, fertilization.
- Human reproduction: male and female reproductive systems, menstrual cycle, reproductive health and contraception.

Unit 4: Heredity (3 Marks)

- Accumulation of variation, Mendel's laws of inheritance (monohybrid and dihybrid crosses).
- Mechanism of sex determination in human beings.

Unit 5: Our Environment (5 Marks)

- Ecosystem and its components; trophic levels, food chains, and food webs.
- Biological magnification; ozone layer depletion and its causes.
- Solid waste management and biodegradable vs non-biodegradable wastes.

Prescribed Textbooks:

- A Textbook of Science for Class X Published by JKBOSE / NCERT